Module 6: Blockchain 3.0 [5L]:

Hyperledger fabric, the plug and play platform and mechanisms in permissioned blockchain.

Hyperledger fabric

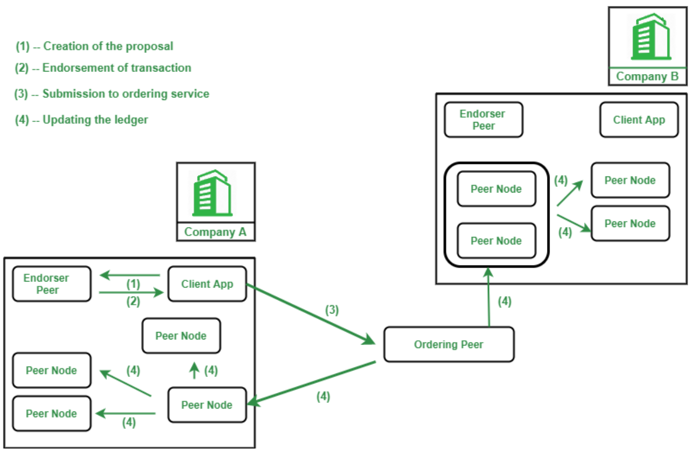
Hyperledger Fabric is designed for use in enterprise-level applications, and it is characterized by its modular architecture, permissioned network, and smart contract functionality, known as "chaincode".

- The platform provides a high degree of security, privacy, and scalability, and it supports the development of custom blockchain solutions for various use cases across industries such as finance, supply chain, and healthcare.
- Hyperledger Fabric operates as a network of nodes, where each node performs a specific function, such as validating transactions, maintaining the ledger, and executing chaincode.
- Transactions are validated and ordered by a consensus mechanism, which ensures the integrity and consistency of the ledger.

How does Hyperledger Fabric Work?

For each and every transaction in the fabric, the following steps are followed-

1. Creation of the proposal: Imagine a deal between a smartphone manufacturer company and a smartphone dealership. The transaction begins when a member organization proposes or invokes a transaction request with the help of the client application or portal. Then the client application sends the proposal to peers in each organization for endorsement.
2. Endorsement of the transaction: After the proposal reaches the endorser peers (peers in each organization for endorsement of a proposal) the peer checks the fabric certificate authority of the requesting member and other details that are needed to authenticate the transaction. Then it executes the chain code and returns a response. This response indicates the approval or rejection of the following transaction. The response is carried out to the client.
3. Submission to ordering service: After receiving the endorsement output, the approved transactions are sent to the ordering service by the client-side application. The peer responsible for the ordering service includes the transaction into a specific block and sends it to the peer nodes of different members of the network.
4. Updating the ledger: After receiving this block the peer nodes of such organizations update their local ledger with this block. Hence the new transactions are now committed.



Industry Use Cases For Hyperledger Fabric

1. Supply Chain: Supply chains are global or regional webs of suppliers, manufacturers, and retailers of a particular product. Hyperledger Fabric networks can improve the transaction processes of the supply chain by increasing the clarity and traceability of transactions within the fabric. On a Fabric network, enterprises having authentication to access the ledger can view the data of the previous transactions. This fact increases accountability and reduces the risk of counterfeiting of the transactions. Real-time production and shipping updates can be updated to the ledger. Which can help us to track the product condition in a much faster, simpler, and efficient way.

2. Trading and Asset Transfer: Trading and asset transfer requires many organizations or members like importers, exporters, banks, brokers. They work with one another. And even in the era of digitalization a lot of paperwork is going on in this sector. But using Hyperledger they can transact and interact with each other in a paperless way. The Hyperledger fabric can add the same layer of trust as the document signed by a trusted authority. This also increases the performance of the system.
Another benefit of Hyperledger fabric is that assets can be dematerialized on the blockchain network with the help of Hyperledger fabric. Due to this traders or stakeholders will be able to have direct access to their financial securities and they can trade it anytime.

3. Insurance: The insurance industry spends billions to avoid insurance frauds or falsified claims. With the help of Hyperledger fabric, the Insurance company can refer to the transaction data that is stored inside the ledger. Hyperledger Fabric can also make the processing of claims faster using the chain code and automate the payment. This process will be also helpful for multi-party subrogation claims processing. Where it can automate repayment from the fault party back to the insurance company. Verification of identity or KYC process will be easy using this private blockchain.

Benefits Of Hyperledger Fabric

1. Open Source: Hyperledger fabric is an open-source blockchain framework hosted by the Linux foundation. It has an active community of developers The code is designed to be publicly accessible. Anyone in the community can see, modify, and distribute the code as they see fit. People across the world can come and help to develop the source code.

2. Private and Confidential: In a public blockchain network each and every node in the network is receiving a copy of the whole ledger. Thus keeping privacy becomes a much bigger concern as everything is open to everyone. In addition to this one, the identities of all the participating members are not known and authenticated. Anyone can participate as it is a public blockchain. But in the case of Hyperledger fabric, the identities of all participating members are authenticated. And the ledger is only exposed to the authenticated members. This benefit is the most useful in industry-level cases, like banking, insurance, etc where customer data should be kept private.

3. Access Control: In the Hyperledger fabric, there is a virtual blockchain network on top of the physical blockchain network. It has its own access rules. It employs its own mechanism for transaction ordering and provides an additional layer of access control. It is especially useful when members want to limit the exposure of data and make it private. Such that it can be viewed by the related parties only. As an example when two competitors are on the same network. The fabric also offers private data collection and accessibility, where one competitor can control the access to its own data such that the data do not get exposed to the other competitor.

4. Chaincode Functionality: It includes a container technology to host smart contracts called *chain code* that defines the business rules of the system. And it's designed to support various pluggable components and to accommodate the complexity that exists across the entire economy. This is useful for some of the specific types of transactions like asset ownership change.

5. Performance: As the Hyperledger fabric is a private blockchain network, There is no need to validate the transactions on this network so the transaction speed is faster, resulting in a better performance.

Limitation of Hyperledger Fabric

Hyperledger Fabric is a robust and flexible platform for developing blockchain applications, but like any technology, it has certain limitations:

1. **Scalability:** Hyperledger Fabric is designed for permissioned networks, where the participants are known and trusted, which can limit its scalability for large-scale public networks.
2. **Performance:** The performance of Hyperledger Fabric can be impacted by factors such as network size, network configuration, and the complexity of chaincode, which can limit its ability to handle high volumes of transactions.
3. **Complexity:** Setting up and configuring a Hyperledger Fabric network can be complex, requiring a deep understanding of the technology and its components.
4. **Compatibility:** Hyperledger Fabric is designed to be used with specific programming languages, such as Go and JavaScript, which can limit its compatibility with other technologies and programming languages.
5. **Cost:** Running a Hyperledger Fabric network requires infrastructure and resources, which can add costs to the deployment and operation of blockchain applications.
6. **Interoperability:** Hyperledger Fabric is designed to be used within a single network, and its interoperability with other blockchain platforms is limited.

The plug and play platform and mechanisms in permissioned blockchain

Hyperledger Fabric is the primary "plug and play" permissioned blockchain platform, utilizing a modular architecture that allows interchangeable components for consensus, membership services, and cryptography. It offers pluggable mechanisms, including tailored consensus algorithms (e.g., Raft, Kafka, BFT) and identity management (MSP), designed for enterprise privacy and high-performance scalability.

Key Plug and Play Mechanisms in Permissioned Blockchains

- **Modular Architecture:** Platforms like Fabric allow developers to choose components, such as substituting the consensus protocol, without altering the underlying ledger structure.
- **Consensus Algorithms:** Unlike energy-intensive proof-of-work, permissioned networks use plug-and-play consensus models, such as PBFT (Practical Byzantine Fault Tolerance) for high-trust environments, or leader-based protocols (e.g., Raft, Kafka) for speed.
- **Membership Service Provider (MSP):** A pluggable component that handles user authentication and issuance of certificates, ensuring only verified participants join. [1, 2]

- **Smart Contract ("Chaincode") Engines:** These platforms allow modular smart contract execution, often supporting multiple programming languages (Go, Java, Node.js) in isolation.
- **Privacy Components:** Channels (in Fabric) and private data collections can be plugged in to create private "sub-networks" within the larger consortium, restricting data visibility to specific participants.